

New calculation of quadrupole setting for Hall A Möller Polarimeter

1 Old Setting

The last calculations of Möller Polarimeter Quadrupole Setting were done at the beginning 1998. The angle acceptance calculations with TRANSPORT code were used at the ground of these Settings. Transport results were tested and corrected at some points at the energy scale and as result some setting curves were produced for quick overhead of Moller quads settings.

After experimental investigations of quads setting by last quads current scanning we found that calculated setting is placed too closely to the apperture limitations and that the most effective acceptance condition are realized with additional focusing field of the third quad. Some correction to the setting curves were done and calculated value were directed to accelerator division. It was proved to be that it is possible to use the calculated Moller setting for the main beam tuning. It was very conveniently to have the same setting for both tunings.

2 New approach

However some discrepancies between the proposed quads setting and effective Moller acceptance at the small energies were found. So the setting calculations were repeated. In the ground these calculations the new codes written for UNICS machines were used. The calculations are based on comfile *tst.com* (see Appendix 1) including some fortran codes. At these codes the random quads setting at the definite quads fields range was created (see code *varsetn.f* at Appendix 2). For this random setting the special idt-file for Raytrace calculation were created (see code *setqd.f* at Appendix 3 and idt-file *x.idt* at Appendix 4). The kinematic parameters of the Moller electrons rays were chosen by code *setqd.f* starting from $\theta^*=90^\circ$ at the CM system as follows:

$$\theta_{nk}^* = 90^\circ + (-1)^k \times 1^\circ \times (n - 1)$$

where n is changed at the range from 1 to 50 and k doubles the angle range to both sides relatively $\theta^*=90^\circ$. The kinematics formula transform the CM system values to corresponding values at Lab-system. For simplicity of calculation the φ angle was equal to zero. Effects of the multiple scattering the primary and secondary electrons and Levchuk-effect are neglected in these calculation. A final idt-file *x.idt* with the random quads settings and a set of 100 Moller rays was calculated by using of RAYTRACE code *ray91 < x.idt > x.out*.

After Raytrace calculations the angle acceptance value was done (see code *setdp.f* at Appendix 5) by number of unstopped rays, starting from 90° to the first stopped ray. The wrong hitting to 90° point is estimated as zero acceptance.

The calculation results are represented as three scattered plots for each quads - quads fields versus θ^* -angle acceptance. The example the calculated scattered plots H%dTh and for each quads is shown at fig.1(a-c). For estimation of the best quads setting some plots (see fig.1(d-f) for $N_{hit} = f(H_{Qd})$ are drawn with additional angular cuts $Cut = 2\theta^*$ and mean values the best

field values for each quads were calculated. However the *RMS*-value for the quads field too high. This fact reflects at first the small statistics for calculated results and at second the sufficiently wide range for field values for each of quads. The additional information is done at the $N_{hit} = f(H_{sum})$ plot, where H_{sum} is summary field all of three quads. We can see at fig.2 that the single best setting is present for maximum θ^* -angle acceptance value.

The final results of these calculations for full energy range from 1.0 GeV to 4.2 GeV are presented at the fig.3(a-c). It can be seen that the settings curves show the irregular behaviour from point to point as results mentioned poor statistic and can be reasonable smoothed.

Unfortunately the sum value for all three quads field values for the best acceptance setting, that presented at fig.4(a), shows the very smooth behaviour and very small statistic errors near 1-2%. It supplies the simple procedure to fix the best quads field setting without additional long-term calculations: we fix the field values, for example, in the second and the third quads by using very simple linear approximations. The needed value for the first quads is result, calculated from additional acceptance calculation at above mentioned procedure with using simplified code *varsetns.f* (see Appendix 6). The result of these calculation is shown at fig.5.

The result for angular acceptance value for different energy and optimal quads field setting is shown at fig.4(b). Of course, this acceptance reflects the angular acceptance for Moller events in coincidence just after exit from dipole gaps and is a result of the quads kinematic focusing. The dipole dispersion at the detector plane, equal to near 1cm/% restricts the momentum (angular) acceptance the value $\pm 14\%$ ($\pm 15^\circ$ at c.m.s.).

The final results for quads setting for full energy range are listed at the Table 1.

Table 1. Moller Settings for Quads and dipole field values.

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
800	1.282	0.150	0.200	1.632	1.418
810	1.295	0.150	0.206	1.651	1.435
820	1.307	0.150	0.213	1.670	1.453
830	1.319	0.150	0.219	1.688	1.471
840	1.331	0.150	0.226	1.707	1.488
850	1.343	0.150	0.232	1.725	1.506
860	1.355	0.150	0.238	1.744	1.524
870	1.367	0.150	0.245	1.762	1.542
880	1.379	0.150	0.251	1.780	1.559
890	1.391	0.150	0.257	1.798	1.577
900	1.402	0.150	0.264	1.816	1.595
910	1.412	0.150	0.270	1.832	1.613
920	1.423	0.150	0.277	1.850	1.630
930	1.433	0.150	0.283	1.866	1.648
940	1.443	0.150	0.289	1.882	1.666
950	1.453	0.150	0.296	1.899	1.683
960	1.463	0.150	0.302	1.915	1.701

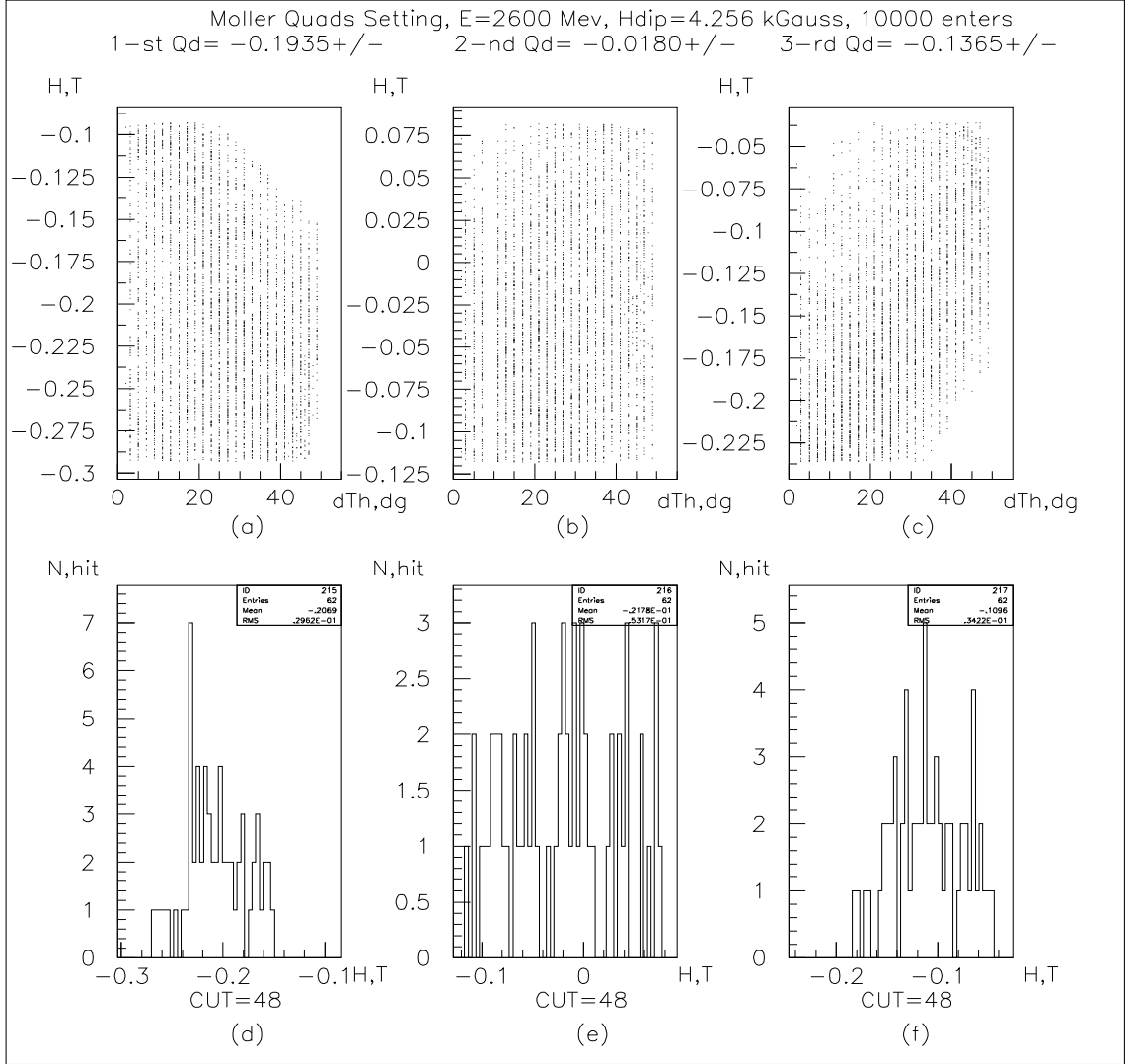


Figure 1: The plots for free drawing ranges of each Quads fields.

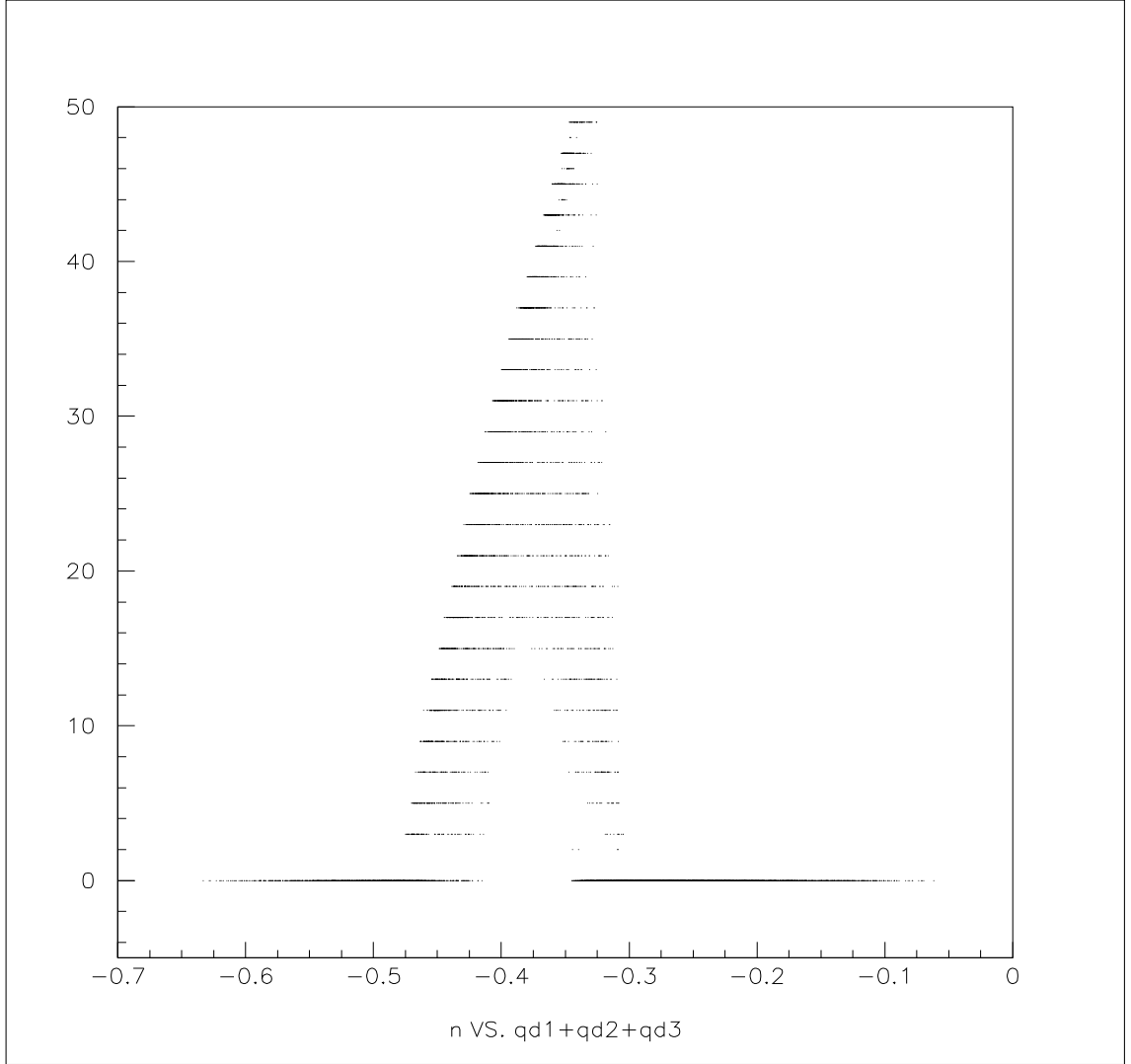


Figure 2: The plots for angular acceptance in dependence from fields sum.

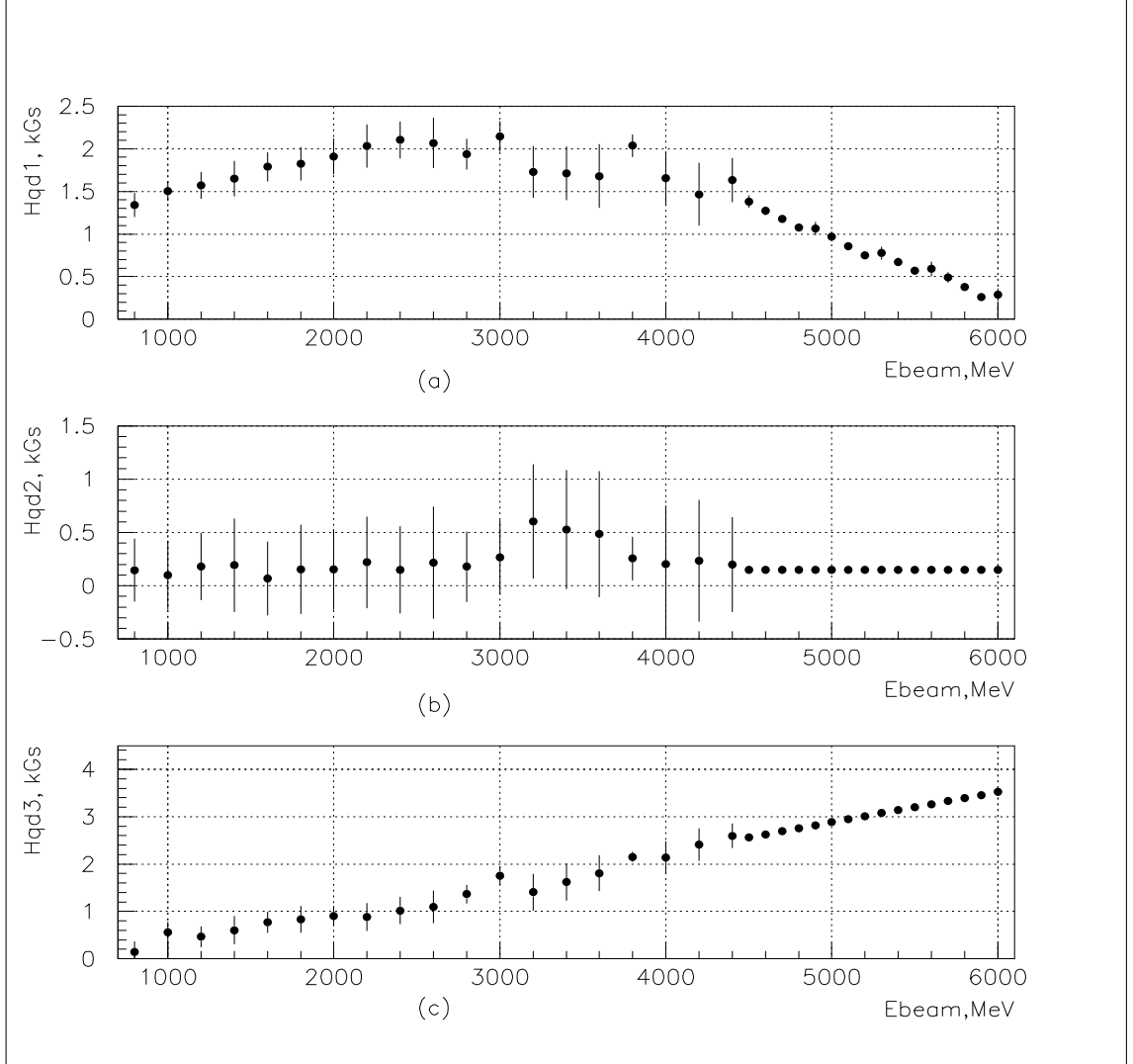


Figure 3: The Quads Setting for free drawing ranges of each Quads fields (for points above 4400 MeV the second and third quads fields are fixed).

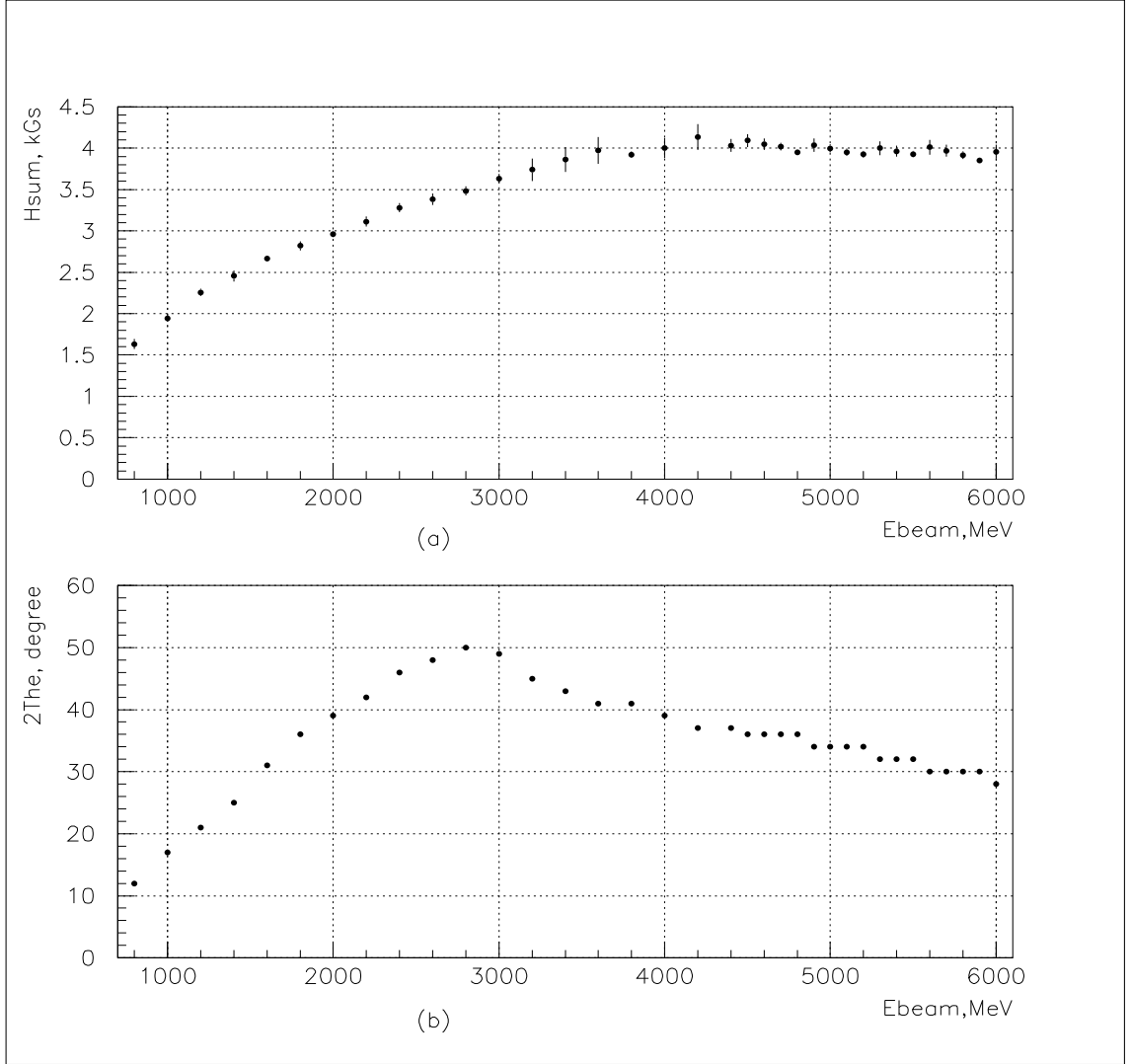


Figure 4: The summary three quads field and θ^* -angle acceptance.

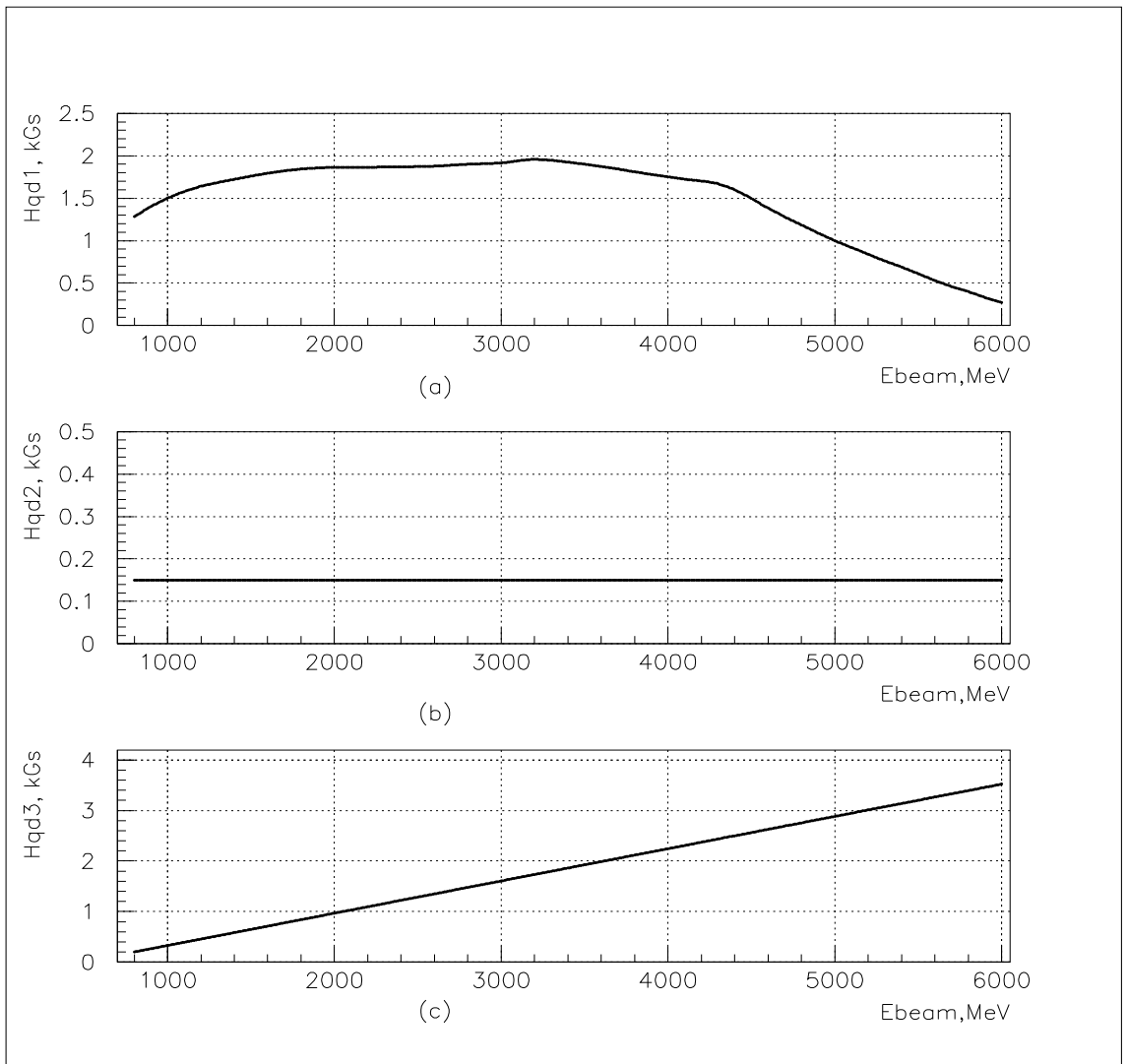


Figure 5: The final Quads Setting.

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
970	1.472	0.150	0.309	1.931	1.719
980	1.482	0.150	0.315	1.947	1.737
990	1.492	0.150	0.321	1.963	1.754
1000	1.502	0.150	0.328	1.980	1.772
1010	1.509	0.150	0.334	1.993	1.790
1020	1.517	0.150	0.341	2.008	1.807
1030	1.525	0.150	0.347	2.022	1.825
1040	1.534	0.150	0.353	2.037	1.843
1050	1.542	0.150	0.360	2.052	1.861
1060	1.549	0.150	0.366	2.065	1.878
1070	1.557	0.150	0.373	2.080	1.896
1080	1.565	0.150	0.379	2.094	1.914
1090	1.572	0.150	0.385	2.107	1.931
1100	1.579	0.150	0.392	2.121	1.949
1110	1.586	0.150	0.398	2.134	1.967
1120	1.592	0.150	0.405	2.147	1.985
1130	1.599	0.150	0.411	2.160	2.002
1140	1.605	0.150	0.418	2.173	2.020
1150	1.611	0.150	0.424	2.185	2.038
1160	1.617	0.150	0.431	2.198	2.056
1170	1.624	0.150	0.437	2.211	2.073
1180	1.630	0.150	0.443	2.223	2.091
1190	1.636	0.150	0.450	2.236	2.109
1200	1.642	0.150	0.456	2.248	2.126
1210	1.644	0.150	0.462	2.256	2.144
1220	1.649	0.150	0.469	2.268	2.162
1230	1.654	0.150	0.475	2.279	2.180
1240	1.659	0.150	0.482	2.291	2.197
1250	1.663	0.150	0.488	2.301	2.215
1260	1.667	0.150	0.495	2.312	2.233
1270	1.672	0.150	0.501	2.323	2.250
1280	1.676	0.150	0.507	2.333	2.268
1290	1.680	0.150	0.513	2.343	2.286
1300	1.684	0.150	0.519	2.353	2.304
1310	1.688	0.150	0.525	2.363	2.321
1320	1.692	0.150	0.532	2.374	2.339
1330	1.696	0.150	0.538	2.384	2.357
1340	1.700	0.150	0.545	2.395	2.374
1350	1.704	0.150	0.551	2.405	2.392

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
1360	1.708	0.150	0.558	2.416	2.410
1370	1.712	0.150	0.564	2.426	2.427
1380	1.716	0.150	0.570	2.436	2.445
1390	1.719	0.150	0.576	2.445	2.463
1400	1.723	0.150	0.583	2.456	2.481
1410	1.727	0.150	0.589	2.466	2.499
1420	1.731	0.150	0.596	2.477	2.516
1430	1.735	0.150	0.602	2.487	2.534
1440	1.739	0.150	0.609	2.498	2.552
1450	1.742	0.150	0.615	2.507	2.569
1460	1.746	0.150	0.622	2.518	2.587
1470	1.750	0.150	0.628	2.528	2.605
1480	1.753	0.150	0.634	2.537	2.623
1490	1.757	0.150	0.641	2.548	2.640
1500	1.761	0.150	0.647	2.558	2.658
1510	1.764	0.150	0.653	2.567	2.676
1520	1.768	0.150	0.660	2.578	2.693
1530	1.771	0.150	0.666	2.588	2.711
1540	1.775	0.150	0.673	2.598	2.729
1550	1.778	0.150	0.679	2.607	2.747
1560	1.781	0.150	0.685	2.616	2.764
1570	1.785	0.150	0.692	2.627	2.782
1580	1.788	0.150	0.698	2.636	2.800
1590	1.791	0.150	0.705	2.646	2.817
1600	1.795	0.150	0.711	2.656	2.835
1610	1.798	0.150	0.717	2.665	2.853
1620	1.801	0.150	0.724	2.675	2.871
1630	1.804	0.150	0.730	2.684	2.888
1640	1.807	0.150	0.737	2.694	2.906
1650	1.809	0.150	0.743	2.702	2.924
1660	1.812	0.150	0.750	2.712	2.942
1670	1.815	0.150	0.756	2.721	2.959
1680	1.817	0.150	0.763	2.730	2.077
1690	1.820	0.150	0.769	2.739	2.995
1700	1.822	0.150	0.776	2.748	3.012
1710	1.825	0.150	0.782	2.757	3.030
1720	1.827	0.150	0.789	2.766	3.048
1730	1.830	0.150	0.795	2.775	3.066

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
1740	1.832	0.150	0.802	2.784	3.084
1750	1.834	0.150	0.808	2.792	3.101
1760	1.836	0.150	0.815	2.801	3.119
1770	1.838	0.150	0.822	2.810	3.136
1780	1.840	0.150	0.828	2.818	3.154
1790	1.842	0.150	0.834	2.826	3.172
1800	1.844	0.150	0.839	2.833	3.190
1810	1.846	0.150	0.845	2.841	3.207
1820	1.847	0.150	0.851	2.848	3.225
1830	1.848	0.150	0.858	2.856	3.243
1840	1.850	0.150	0.864	2.864	3.260
1850	1.851	0.150	0.871	2.872	3.278
1860	1.852	0.150	0.877	2.879	3.296
1870	1.853	0.150	0.883	2.886	3.314
1880	1.854	0.150	0.890	2.894	3.331
1890	1.855	0.150	0.896	2.901	3.349
1900	1.856	0.150	0.903	2.909	3.367
1910	1.857	0.150	0.909	2.916	3.385
1920	1.858	0.150	0.916	2.924	3.402
1930	1.859	0.150	0.922	2.931	3.420
1940	1.860	0.150	0.929	2.939	3.438
1950	1.861	0.150	0.935	2.946	3.455
1960	1.862	0.150	0.942	2.954	3.473
1970	1.863	0.150	0.948	2.961	3.491
1980	1.864	0.150	0.955	2.969	3.509
1990	1.864	0.150	0.961	2.975	3.526
2000	1.864	0.150	0.967	2.981	3.544
2010	1.864	0.150	0.973	2.987	3.562
2020	1.864	0.150	0.980	2.994	3.579
2030	1.865	0.150	0.986	3.001	3.597
2040	1.865	0.150	0.993	3.008	3.615
2050	1.865	0.150	0.999	3.014	3.633
2060	1.865	0.150	1.006	3.021	3.650
2070	1.865	0.150	1.012	3.027	3.668
2080	1.865	0.150	1.018	3.033	3.686
2090	1.865	0.150	1.024	3.039	3.704
2100	1.865	0.150	1.030	3.045	3.721
2110	1.865	0.150	1.036	3.051	3.739
2120	1.865	0.150	1.043	3.058	3.757

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
2130	1.865	0.150	1.049	3.064	3.774
2140	1.865	0.150	1.056	3.071	3.792
2150	1.866	0.150	1.062	3.078	3.810
2160	1.866	0.150	1.069	3.085	3.828
2170	1.866	0.150	1.075	3.091	3.845
2180	1.866	0.150	1.081	3.097	3.863
2190	1.866	0.150	1.088	3.104	3.881
2200	1.866	0.150	1.094	3.110	3.898
2210	1.866	0.150	1.100	3.116	3.916
2220	1.866	0.150	1.107	3.123	3.934
2230	1.866	0.150	1.113	3.129	3.952
2240	1.866	0.150	1.120	3.126	3.969
2250	1.866	0.150	1.126	3.132	3.987
2260	1.867	0.150	1.133	3.140	4.005
2270	1.867	0.150	1.140	3.147	4.022
2280	1.867	0.150	1.146	3.153	4.040
2290	1.867	0.150	1.152	3.159	4.058
2300	1.867	0.150	1.158	3.175	4.076
2310	1.867	0.150	1.164	3.181	4.093
2320	1.867	0.150	1.171	3.188	4.111
2330	1.867	0.150	1.177	3.194	4.129
2340	1.867	0.150	1.184	3.201	4.147
2350	1.868	0.150	1.190	3.208	4.164
2360	1.868	0.150	1.197	3.215	4.182
2370	1.868	0.150	1.203	3.221	4.200
2380	1.868	0.150	1.209	3.227	4.217
2390	1.868	0.150	1.216	3.234	4.235
2400	1.868	0.150	1.222	3.240	4.253
2410	1.868	0.150	1.228	3.246	4.271
2420	1.869	0.150	1.235	3.254	4.288
2430	1.869	0.150	1.241	3.260	4.306
2440	1.870	0.150	1.248	3.268	4.324
2450	1.870	0.150	1.254	3.274	4.341
2460	1.871	0.150	1.261	3.282	4.359
2470	1.871	0.150	1.267	3.288	4.377
2480	1.872	0.150	1.274	3.296	4.395
2490	1.872	0.150	1.280	3.302	4.412
2500	1.873	0.150	1.286	3.309	4.430

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
2510	1.873	0.150	1.292	3.315	4.448
2520	1.874	0.150	1.299	3.323	4.465
2530	1.874	0.150	1.305	3.329	4.483
2540	1.875	0.150	1.312	3.337	4.501
2550	1.875	0.150	1.318	3.343	4.519
2560	1.876	0.150	1.325	3.351	4.536
2570	1.876	0.150	1.331	3.357	4.554
2580	1.877	0.150	1.337	3.364	4.572
2590	1.877	0.150	1.343	3.370	4.590
2600	1.878	0.150	1.350	3.378	4.607
2610	1.879	0.150	1.356	3.387	4.625
2620	1.880	0.150	1.363	3.393	4.643
2630	1.881	0.150	1.369	3.400	4.660
2640	1.882	0.150	1.376	3.408	4.678
2650	1.883	0.150	1.382	3.415	4.696
2660	1.884	0.150	1.389	3.423	4.714
2670	1.886	0.150	1.395	3.431	4.731
2680	1.887	0.150	1.402	3.439	4.749
2690	1.888	0.150	1.408	3.446	4.767
2700	1.889	0.150	1.414	3.453	4.784
2710	1.890	0.150	1.421	3.461	4.802
2720	1.891	0.150	1.427	3.468	4.820
2730	1.892	0.150	1.434	3.476	4.838
2740	1.893	0.150	1.440	3.483	4.855
2750	1.895	0.150	1.447	3.492	4.873
2760	1.896	0.150	1.453	3.499	4.891
2770	1.897	0.150	1.460	3.507	4.908
2780	1.898	0.150	1.466	3.514	4.926
2790	1.899	0.150	1.472	3.521	4.944
2800	1.900	0.150	1.478	3.528	4.962
2810	1.901	0.150	1.484	3.535	4.979
2820	1.901	0.150	1.491	3.542	4.997
2830	1.902	0.150	1.497	3.549	5.015
2840	1.903	0.150	1.504	3.555	5.033
2850	1.903	0.150	1.510	3.563	5.050
2860	1.904	0.150	1.517	3.571	5.068
2870	1.905	0.150	1.523	3.578	5.086
2880	1.906	0.150	1.529	3.585	5.103

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
2890	1.907	0.150	1.535	3.592	5.121
2900	1.907	0.150	1.542	3.599	5.139
2910	1.908	0.150	1.548	3.606	5.157
2920	1.909	0.150	1.555	3.614	5.174
2930	1.910	0.150	1.561	3.621	5.192
2940	1.911	0.150	1.568	3.629	5.210
2950	1.911	0.150	1.574	3.635	5.227
2960	1.912	0.150	1.581	3.643	5.245
2970	1.913	0.150	1.587	3.650	5.263
2980	1.914	0.150	1.593	3.657	5.281
2990	1.915	0.150	1.599	3.664	5.298
3000	1.915	0.150	1.606	3.671	5.316
3010	1.918	0.150	1.612	3.680	5.334
3020	1.920	0.150	1.619	3.689	5.351
3030	1.922	0.150	1.625	3.697	5.369
3040	1.924	0.150	1.632	3.706	5.387
3050	1.927	0.150	1.638	3.715	5.405
3060	1.930	0.150	1.645	3.725	5.422
3070	1.933	0.150	1.651	3.734	5.440
3080	1.936	0.150	1.658	3.744	5.458
3090	1.939	0.150	1.664	3.753	5.476
3100	1.941	0.150	1.670	3.761	5.493
3110	1.943	0.150	1.676	3.769	5.511
3120	1.945	0.150	1.683	3.778	5.529
3130	1.948	0.150	1.689	3.786	5.546
3140	1.950	0.150	1.696	3.796	5.564
3150	1.952	0.150	1.702	3.804	5.581
3160	1.954	0.150	1.709	3.813	5.600
3170	1.956	0.150	1.715	3.821	5.617
3180	1.957	0.150	1.721	3.828	5.635
3190	1.958	0.150	1.727	3.835	5.653
3200	1.958	0.150	1.733	3.803	5.670
3210	1.958	0.150	1.739	3.847	5.688
3220	1.957	0.150	1.746	3.853	5.706
3230	1.956	0.150	1.752	3.858	5.724
3240	1.955	0.150	1.759	3.864	5.741
3250	1.954	0.150	1.765	3.869	5.759
3260	1.953	0.150	1.772	3.875	5.777

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
3270	1.951	0.150	1.778	3.879	5.794
3280	1.950	0.150	1.785	3.885	5.812
3290	1.948	0.150	1.791	3.889	5.830
3300	1.947	0.150	1.797	3.894	5.848
3310	1.945	0.150	1.803	3.898	5.865
3320	1.942	0.150	1.810	3.902	5.883
3330	1.940	0.150	1.816	3.906	5.901
3340	1.938	0.150	1.823	3.911	5.919
3350	1.936	0.150	1.829	3.915	5.936
3360	1.934	0.150	1.836	3.920	5.954
3370	1.932	0.150	1.842	3.924	5.972
3380	1.929	0.150	1.849	3.928	5.989
3390	1.927	0.150	1.855	3.932	6.007
3400	1.923	0.150	1.861	3.934	6.023
3410	1.923	0.150	1.867	3.940	6.043
3420	1.920	0.150	1.874	3.944	6.060
3430	1.918	0.150	1.880	3.948	6.078
3440	1.916	0.150	1.887	3.953	6.096
3450	1.914	0.150	1.893	3.957	6.113
3460	1.911	0.150	1.900	3.961	6.131
3470	1.908	0.150	1.906	3.964	6.149
3480	1.906	0.150	1.913	3.969	6.167
3490	1.904	0.150	1.919	3.973	6.184
3500	1.901	0.150	1.925	3.976	6.202
3510	1.898	0.150	1.931	3.979	6.220
3520	1.896	0.150	1.938	3.984	6.237
3530	1.894	0.150	1.944	3.988	6.255
3540	1.891	0.150	1.951	3.992	6.273
3550	1.888	0.150	1.957	3.995	6.291
3560	1.885	0.150	1.963	3.998	6.308
3570	1.882	0.150	1.970	4.002	6.326
3580	1.879	0.150	1.976	4.005	6.344
3590	1.876	0.150	1.982	4.008	6.362
3600	1.876	0.150	1.989	4.017	6.379
3610	1.871	0.150	1.995	4.016	6.397
3620	1.868	0.150	2.002	4.020	6.415
3630	1.865	0.150	2.008	4.023	6.432
3650	1.859	0.150	2.021	4.030	6.468
3660	1.857	0.150	2.028	4.035	6.486

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
3660	1.857	0.150	2.028	4.035	6.486
3670	1.854	0.150	2.034	4.038	6.503
3680	1.850	0.150	2.041	4.041	6.521
3690	1.848	0.150	2.047	4.045	6.539
3700	1.845	0.150	2.053	4.048	6.556
3710	1.841	0.150	2.059	4.050	6.574
3720	1.838	0.150	2.066	4.054	6.592
3730	1.835	0.150	2.072	4.057	6.610
3740	1.832	0.150	2.079	4.061	6.627
3750	1.829	0.150	2.085	4.064	6.645
3760	1.825	0.150	2.092	4.067	6.663
3770	1.821	0.150	2.098	4.069	6.680
3780	1.818	0.150	2.104	4.072	6.698
3790	1.815	0.150	2.110	4.075	6.716
3800	1.813	0.150	2.117	4.080	6.734
3810	1.810	0.150	2.123	4.083	6.751
3820	1.807	0.150	2.130	4.087	6.769
3830	1.803	0.150	2.136	4.089	6.787
3840	1.799	0.150	2.143	4.092	6.805
3850	1.796	0.150	2.149	4.095	6.822
3860	1.793	0.150	2.156	4.099	6.840
3870	1.790	0.150	2.162	4.102	6.858
3880	1.787	0.150	2.169	4.106	6.975
3890	1.784	0.150	2.175	4.109	6.893
3900	1.781	0.150	2.181	4.112	6.911
3910	1.778	0.150	2.187	4.115	6.929
3920	1.775	0.150	2.194	4.119	6.946
3930	1.772	0.150	2.200	4.122	6.064
3940	1.769	0.150	2.207	4.126	6.982
3950	1.766	0.150	2.213	4.129	6.999
3960	1.763	0.150	2.220	4.133	7.017
3970	1.760	0.150	2.226	4.136	7.035
3980	1.757	0.150	2.232	4.139	7.053
3990	1.754	0.150	2.238	4.142	7.070
4000	1.742	0.150	2.244	4.136	7.088
4010	1.749	0.150	2.250	4.149	7.106
4020	1.746	0.150	2.257	4.153	7.123
4030	1.743	0.150	2.263	4.156	7.141
4040	1.740	0.150	2.270	4.160	7.159

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
4050	1.738	0.150	2.276	4.164	7.177
4060	1.735	0.150	2.283	4.168	7.194
4070	1.733	0.150	2.289	4.172	7.212
4080	1.730	0.150	2.296	4.176	7.230
4090	1.728	0.150	2.302	4.180	7.248
4100	1.725	0.150	2.308	4.183	7.265
4110	1.723	0.150	2.314	4.187	7.283
4120	1.720	0.150	2.321	4.191	7.301
4130	1.719	0.150	2.327	4.196	7.318
4140	1.716	0.150	2.334	4.200	7.336
4150	1.714	0.150	2.340	4.204	7.354
4160	1.711	0.150	2.347	4.208	7.372
4170	1.709	0.150	2.353	4.212	7.389
4180	1.707	0.150	2.360	4.217	7.407
4190	1.705	0.150	2.366	4.221	7.425
4200	1.705	0.150	2.372	4.227	7.442
4210	1.702	0.150	2.378	4.230	7.460
4220	1.700	0.150	2.385	4.235	7.478
4230	1.696	0.150	2.391	4.237	7.496
4240	1.693	0.150	2.398	4.241	7.513
4250	1.689	0.150	2.404	4.243	7.531
4260	1.686	0.150	2.411	4.247	7.549
4270	1.683	0.150	2.417	4.250	7.566
4280	1.680	0.150	2.424	4.254	7.584
4290	1.676	0.150	2.431	4.257	7.602
4300	1.670	0.150	2.437	4.257	7.620
4310	1.663	0.150	2.443	4.256	7.637
4320	1.657	0.150	2.450	4.257	7.655
4330	1.651	0.150	2.456	4.257	7.673
4340	1.646	0.150	2.463	4.259	7.691
4350	1.639	0.150	2.469	4.258	7.708
4360	1.631	0.150	2.476	4.257	7.726
4370	1.624	0.150	2.482	4.256	7.744
4380	1.617	0.150	2.488	4.255	7.761
4390	1.608	0.150	2.492	4.250	7.779
4400	1.600	0.150	2.500	4.250	7.797
4410	1.589	0.150	2.506	4.244	7.815
4420	1.579	0.150	2.513	4.242	7.832

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
4430	1.569	0.150	2.519	4.238	7.850
4440	1.559	0.150	2.525	4.234	7.868
4450	1.549	0.150	2.531	4.230	7.886
4460	1.539	0.150	2.538	4.227	7.903
4470	1.529	0.150	2.545	4.224	7.921
4480	1.519	0.150	2.551	4.220	7.939
4490	1.509	0.150	2.557	4.216	7.956
4500	1.500	0.150	2.563	4.213	7.974
4510	1.488	0.150	2.570	4.208	7.920
4520	1.476	0.150	2.576	4.202	7.938
4530	1.464	0.150	2.583	4.197	7.955
4540	1.452	0.150	2.590	4.192	7.973
4550	1.440	0.150	2.596	4.186	7.991
4560	1.427	0.150	2.603	4.180	8.008
4570	1.415	0.150	2.609	4.174	8.026
4580	1.405	0.150	2.616	4.171	8.043
4590	1.395	0.150	2.622	4.167	8.061
4600	1.385	0.150	2.628	4.163	8.078
4610	1.375	0.150	2.635	4.160	8.096
4620	1.364	0.150	2.641	4.155	8.114
4630	1.354	0.150	2.648	4.152	8.131
4640	1.343	0.150	2.654	4.147	8.149
4650	1.333	0.150	2.660	4.143	8.166
4660	1.322	0.150	2.667	4.139	8.184
4670	1.312	0.150	2.673	4.135	8.201
4680	1.301	0.150	2.680	4.131	8.219
4690	1.290	0.150	2.686	4.126	8.236
4700	1.280	0.150	2.692	4.122	8.254
4710	1.270	0.150	2.699	4.119	8.272
4720	1.261	0.150	2.705	4.116	8.289
4730	1.251	0.150	2.711	4.112	8.307
4740	1.242	0.150	2.717	4.109	8.324
4750	1.232	0.150	2.723	4.105	8.342
4760	1.223	0.150	2.730	4.103	8.359
4770	1.213	0.150	2.736	4.099	8.377
4780	1.204	0.150	2.743	4.097	8.394
4790	1.195	0.150	2.749	4.094	8.412
4800	1.185	0.150	2.755	4.090	8.429

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
4810	1.176	0.150	2.762	4.088	8.447
4820	1.166	0.150	2.768	4.084	8.465
4830	1.157	0.150	2.775	4.082	8.482
4840	1.147	0.150	2.781	4.078	8.500
4850	1.138	0.150	2.787	4.075	8.517
4860	1.128	0.150	2.794	4.072	8.535
4870	1.119	0.150	2.800	4.069	8.553
4880	1.109	0.150	2.807	4.066	8.570
4890	1.100	0.150	2.813	4.063	8.588
4900	1.090	0.150	2.819	4.059	8.605
4910	1.081	0.150	2.826	4.057	8.623
4920	1.072	0.150	2.832	4.054	8.640
4930	1.063	0.150	2.839	4.052	8.658
4940	1.054	0.150	2.845	4.049	8.675
4950	1.045	0.150	2.851	4.046	8.693
4960	1.036	0.150	2.858	4.044	8.711
4970	1.027	0.150	2.864	4.041	8.728
4980	1.018	0.150	2.871	4.039	8.746
4990	1.009	0.150	2.877	4.036	8.763
5000	1.000	0.150	2.883	4.033	8.781
5010	0.991	0.150	2.890	4.031	8.798
5020	0.983	0.150	2.896	4.029	8.816
5030	0.975	0.150	2.903	4.028	8.834
5040	0.967	0.150	2.909	4.026	8.851
5050	0.959	0.150	2.915	4.024	8.869
5060	0.951	0.150	2.922	4.023	8.886
5070	0.943	0.150	2.928	4.021	8.904
5080	0.935	0.150	2.935	4.020	8.921
5090	0.927	0.150	2.941	4.018	8.939
5100	0.920	0.150	2.947	4.017	8.956
5110	0.912	0.150	2.954	4.016	8.974
5120	0.904	0.150	2.960	4.014	8.992
5130	0.896	0.150	2.967	4.013	9.009
5140	0.888	0.150	2.973	4.011	9.027
5150	0.880	0.150	2.979	4.009	9.044
5160	0.872	0.150	2.986	4.008	9.062
5170	0.864	0.150	2.992	4.006	9.079
5180	0.856	0.150	2.999	4.005	9.097

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
5190	0.848	0.150	3.005	4.003	9.114
5200	0.840	0.150	3.011	4.001	9.132
5210	0.832	0.150	3.018	4.000	9.150
5220	0.824	0.150	3.024	3.998	9.167
5230	0.816	0.150	3.031	3.997	9.185
5240	0.808	0.150	3.037	3.995	9.202
5250	0.800	0.150	3.043	3.993	9.220
5260	0.792	0.150	3.050	3.992	9.237
5270	0.784	0.150	3.056	3.990	9.255
5280	0.776	0.150	3.063	3.989	9.272
5290	0.768	0.150	3.069	3.987	9.290
5300	0.760	0.150	3.075	3.985	9.308
5310	0.753	0.150	3.082	3.985	9.325
5320	0.746	0.150	3.088	3.984	9.343
5330	0.739	0.150	3.095	3.984	9.360
5340	0.732	0.150	3.101	3.983	9.378
5350	0.725	0.150	3.107	3.982	9.396
5360	0.718	0.150	3.114	3.982	9.413
5370	0.711	0.150	3.120	3.981	9.431
5380	0.704	0.150	3.126	3.980	9.448
5390	0.697	0.150	3.132	3.979	9.466
5400	0.690	0.150	3.138	3.978	9.483
5410	0.682	0.150	3.145	3.977	9.501
5420	0.674	0.150	3.152	3.976	9.518
5430	0.666	0.150	3.158	3.974	9.536
5440	0.658	0.150	3.164	3.972	9.553
5450	0.650	0.150	3.170	3.970	9.571
5460	0.642	0.150	3.177	3.969	9.589
5470	0.634	0.150	3.183	3.967	9.606
5480	0.626	0.150	3.190	3.966	9.624
5490	0.618	0.150	3.196	3.964	9.641
5500	0.610	0.150	3.202	3.962	9.659
5510	0.602	0.150	3.209	3.961	9.676
5520	0.594	0.150	3.215	3.959	9.694
5530	0.586	0.150	3.222	3.958	9.712
5540	0.578	0.150	3.228	3.956	9.729
5550	0.570	0.150	3.234	3.954	9.747
5560	0.562	0.150	3.241	3.953	9.764

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
5570	0.554	0.150	3.247	3.951	9.782
5580	0.546	0.150	3.254	3.949	9.799
5590	0.538	0.150	3.260	3.948	9.817
5600	0.530	0.150	3.266	3.946	9.834
5610	0.523	0.150	3.273	3.946	9.852
5620	0.516	0.150	3.279	3.945	9.870
5630	0.509	0.150	3.286	3.945	9.887
5640	0.502	0.150	3.292	3.944	9.905
5650	0.495	0.150	3.298	3.943	9.922
5660	0.488	0.150	3.305	3.943	9.940
5670	0.481	0.150	3.311	3.942	9.957
5680	0.474	0.150	3.318	3.942	9.975
5690	0.467	0.150	3.324	3.941	9.993
5700	0.460	0.150	3.330	3.940	10.010
5710	0.454	0.150	3.337	3.941	10.028
5720	0.448	0.150	3.343	3.941	10.045
5730	0.442	0.150	3.350	3.942	10.063
5740	0.436	0.150	3.356	3.942	10.080
5750	0.430	0.150	3.362	3.942	10.098
5760	0.424	0.150	3.369	3.943	10.115
5770	0.418	0.150	3.375	3.943	10.133
5780	0.412	0.150	3.382	3.944	10.151
5790	0.406	0.150	3.388	3.944	10.168
5800	0.400	0.150	3.394	3.944	10.186
5810	0.393	0.150	3.401	3.944	10.203
5820	0.386	0.150	3.407	3.943	10.221
5830	0.379	0.150	3.414	3.943	10.238
5840	0.372	0.150	3.420	3.942	10.256
5850	0.365	0.150	3.426	3.941	10.273
5860	0.358	0.150	3.433	3.941	10.291
5870	0.351	0.150	3.440	3.941	10.309
5880	0.344	0.150	3.446	3.940	10.326
5890	0.337	0.150	3.452	3.939	10.344
5900	0.330	0.150	3.458	3.938	10.361
5910	0.324	0.150	3.465	3.939	10.379
5920	0.318	0.150	3.471	3.939	10.396
5930	0.312	0.150	3.477	3.939	10.414
5940	0.306	0.150	3.482	3.938	10.432

Table 1(continue).

E_b, GeV	H_{Qd1}, kGs	H_{Qd2}, kGs	H_{Qd3}, kGs	H_{Sum}, kGs	H_{Dip}, kGs
5950	0.300	0.150	3.489	3.939	10.449
5960	0.294	0.150	3.496	3.940	10.467
5970	0.288	0.150	3.502	3.940	10.484
5980	0.282	0.150	3.509	3.941	10.502
5990	0.276	0.150	3.515	3.941	10.519
6000	0.270	0.150	3.521	3.941	10.537

The quads and dipole setting in BdL-units for full energy range are listed at the Table 2. The used units are Gauss*cm.

Table 2. Moller Settings for Quads and dipole in BdL-values.

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
800	11568	1102	1450	232870
810	11685	1102	1493	235781
820	11794	1102	1544	238692
830	11902	1102	1587	241603
840	12010	1102	1638	244514
850	12119	1102	1682	247425
860	12227	1102	1725	250335
870	12335	1102	1776	253246
880	12443	1102	1820	256157
890	12552	1102	1863	259068
900	12651	1102	1914	261979
910	12741	1102	1957	264890
920	12841	1102	2008	267801
930	12931	1102	2052	270712
940	13021	1102	2095	273622
950	13111	1102	2146	276533
960	13201	1102	2189	279444
970	13283	1102	2240	282355
980	13373	1102	2284	285266
990	13463	1102	2327	288177
1000	13553	1102	2378	291088
1010	13617	1102	2421	293999
1020	13689	1102	2472	296910
1030	13761	1102	2516	299820
1040	13842	1102	2559	302731
1050	13914	1102	2610	305642

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
1060	13978	1102	2653	308553
1070	14050	1102	2704	311465
1080	14122	1102	2748	314375
1090	14185	1102	2791	317286
1100	14248	1102	2842	320197
1110	14311	1102	2885	323108
1120	14366	1102	2936	326018
1130	14429	1102	2980	328929
1140	14483	1102	3030	331840
1150	14537	1102	3074	334751
1160	14591	1102	3125	337662
1170	14654	1102	3168	340573
1180	14708	1102	3212	343484
1190	14763	1102	3262	346395
1200	14817	1102	3306	349305
1210	14835	1102	3349	352216
1220	14880	1102	3400	355127
1230	14925	1102	3444	358038
1240	14970	1102	3494	360949
1250	15006	1102	3538	363860
1260	15042	1102	3589	366771
1270	15087	1102	3632	369682
1280	15124	1102	3676	372593
1290	15160	1102	3719	375503
1300	15196	1102	3763	378414
1310	15232	1102	3806	381325
1320	15268	1102	3857	384236
1330	15304	1102	3901	387147
1340	15340	1102	3951	390058
1350	15376	1102	3995	392969
1360	15412	1102	4046	395880
1370	15448	1102	4089	398790
1380	15485	1102	4133	401701
1390	15512	1102	4176	404612
1400	15548	1102	4227	407523
1410	15584	1102	4270	410434
1420	15620	1102	4321	413345
1430	15656	1102	4365	416256
1440	15692	1102	4415	419167

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
1450	15719	1102	4459	422078
1460	15755	1102	4510	424988
1470	15791	1102	4553	427899
1480	15818	1102	4597	430810
1490	15854	1102	4647	433721
1500	15891	1102	4691	436632
1510	15918	1102	4734	439543
1520	15954	1102	4785	442454
1530	15981	1102	4829	445365
1540	16017	1102	4879	448275
1550	16044	1102	4923	451186
1560	16071	1102	4966	454097
1570	16107	1102	5017	457008
1580	16134	1102	5061	459919
1590	16161	1102	5111	462830
1600	16197	1102	5155	465741
1610	16224	1102	5198	468652
1620	16252	1102	5249	471562
1630	16279	1102	5293	474473
1640	16306	1102	5343	477384
1650	16324	1102	5387	480295
1660	16351	1102	5438	483206
1670	16378	1102	5481	486117
1680	16396	1102	5532	489028
1690	16423	1102	5576	491939
1700	16441	1102	5626	494850
1710	16468	1102	5670	497760
1720	16486	1102	5721	500671
1730	16513	1102	5764	503582
1740	16531	1102	5815	506493
1750	16549	1102	5858	509404
1760	16567	1102	5909	512315
1770	16585	1102	5960	515226
1780	16603	1102	6003	518137
1790	16622	1102	6047	521047
1800	16640	1102	6083	523958
1810	16658	1102	6127	526869
1820	16667	1102	6170	529780
1830	16676	1102	6221	532691

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
1840	16694	1102	6264	535602
1850	16703	1102	6315	538513
1860	16712	1102	6359	541424
1870	16721	1102	6402	544335
1880	16730	1102	6453	547245
1890	16739	1102	6496	550156
1900	16748	1102	6547	553067
1910	16757	1102	6591	555978
1920	16766	1102	6641	558889
1930	16775	1102	6685	561800
1940	16784	1102	6736	564711
1950	16793	1102	6779	567622
1960	16802	1102	6830	570532
1970	16811	1102	6873	573443
1980	16820	1102	6924	576354
1990	16820	1102	6968	579265
2000	16820	1102	7011	582176
2010	16820	1102	7055	585087
2020	16820	1102	7105	587998
2030	16829	1102	7149	590909
2040	16829	1102	7200	593820
2050	16829	1102	7243	596730
2060	16829	1102	7294	599641
2070	16829	1102	7338	602552
2080	16829	1102	7381	605463
2090	16829	1102	7425	608374
2100	16829	1102	7468	611285
2110	16829	1102	7512	614196
2120	16829	1102	7562	617107
2130	16829	1102	7606	620018
2140	16829	1102	7657	622928
2150	16838	1102	7700	625839
2160	16838	1102	7751	628750
2170	16838	1102	7794	631661
2180	16838	1102	7838	634572
2190	16838	1102	7889	637483
2200	16838	1102	7932	640394
2210	16838	1102	7976	643305
2220	16838	1102	8026	646215

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
2230	16838	1102	8070	649126
2240	16838	1102	8121	652037
2250	16838	1102	8164	654948
2260	16847	1102	8215	657859
2270	16847	1102	8266	660770
2280	16847	1102	8309	663681
2290	16847	1102	8353	666592
2300	16847	1102	8396	669503
2310	16847	1102	8440	672413
2320	16847	1102	8490	675324
2330	16847	1102	8534	678235
2340	16847	1102	8585	681146
2350	16856	1102	8628	684057
2360	16856	1102	8679	686968
2370	16856	1102	8722	689879
2380	16856	1102	8766	692790
2390	16856	1102	8817	695700
2400	16856	1102	8860	698611
2410	16856	1102	8904	701522
2420	16865	1102	8954	704433
2430	16865	1102	8998	707344
2440	16874	1102	9049	710255
2450	16874	1102	9092	713166
2460	16883	1102	9143	716077
2470	16883	1102	9187	718988
2480	16892	1102	9237	721898
2490	16892	1102	9281	724809
2500	16901	1102	9324	727720
2510	16901	1102	9368	730631
2520	16910	1102	9419	733542
2530	16910	1102	9462	736453
2540	16919	1102	9513	739364
2550	16919	1102	9556	742275
2560	16928	1102	9607	745185
2570	16928	1102	9651	748096
2580	16937	1102	9694	751007
2590	16937	1102	9738	753918
2600	16946	1102	9788	756829
2610	16955	1102	9832	759740

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
2620	16964	1102	9883	762651
2630	16973	1102	9926	765562
2640	16982	1102	9977	768473
2650	16992	1102	10020	771383
2660	17001	1102	10071	774294
2670	17019	1102	10115	777205
2680	17028	1102	10165	780116
2690	17037	1102	10209	783027
2700	17046	1102	10252	785938
2710	17055	1102	10303	788849
2720	17064	1102	10347	791760
2730	17073	1102	10397	794670
2740	17082	1102	10441	797581
2750	17100	1102	10492	800492
2760	17109	1102	10535	803403
2770	17118	1102	10586	806314
2780	17127	1102	10629	809225
2790	17136	1102	10673	812136
2800	17145	1102	10716	815047
2810	17154	1102	10760	817958
2820	17154	1102	10811	820868
2830	17163	1102	10854	823779
2840	17172	1102	10905	826690
2850	17172	1102	10949	829601
2860	17181	1102	10999	832512
2870	17190	1102	11043	835423
2880	17199	1102	11086	838334
2890	17208	1102	11130	841245
2900	17208	1102	11181	844155
2910	17217	1102	11224	847066
2920	17226	1102	11275	849977
2930	17235	1102	11318	852888
2940	17244	1102	11369	855799
2950	17244	1102	11413	858710
2960	17253	1102	11463	861621
2970	17262	1102	11507	864532
2980	17271	1102	11550	867442
2990	17280	1102	11594	870353
3000	17289	1102	11645	873264

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
3010	17307	1102	11688	876175
3020	17325	1102	11739	879086
3030	17343	1102	11782	881997
3040	17361	1102	11833	884908
3050	17389	1102	11877	887819
3060	17416	1102	11927	890730
3070	17443	1102	11971	893640
3080	17470	1102	12022	896551
3090	17497	1102	12065	899462
3100	17515	1102	12109	902373
3110	17533	1102	12152	905284
3120	17551	1102	12203	908195
3130	17578	1102	12246	911106
3140	17596	1102	12297	914017
3150	17614	1102	12341	916927
3160	17632	1102	12391	919838
3170	17650	1102	12435	922749
3180	17659	1102	12478	925660
3190	17668	1102	12522	928571
3200	17668	1102	12565	931482
3210	17668	1102	12609	934393
3220	17659	1102	12660	937304
3230	17650	1102	12703	940215
3240	17641	1102	12754	943125
3250	17632	1102	12798	946036
3260	17623	1102	12848	948947
3270	17605	1102	12892	951858
3280	17596	1102	12943	954769
3290	17578	1102	12986	957680
3300	17569	1102	13030	960591
3310	17551	1102	13073	963502
3320	17524	1102	13124	966412
3330	17506	1102	13167	969323
3340	17488	1102	13218	972234
3350	17470	1102	13262	975145
3360	17452	1102	13312	978056
3370	17434	1102	13356	980967
3380	17407	1102	13407	983878
3390	17389	1102	13450	986789

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
3400	17371	1102	13494	989700
3410	17352	1102	13537	992610
3420	17325	1102	13588	995521
3430	17307	1102	13631	998432
3440	17289	1102	13682	1001344
3450	17271	1102	13726	1004254
3460	17244	1102	13776	1007165
3470	17217	1102	13820	1010076
3480	17199	1102	13871	1012987
3490	17181	1102	13914	1015898
3500	17154	1102	13958	1018809
3510	17127	1102	14001	1021720
3520	17109	1102	14052	1024631
3530	17091	1102	14095	1027541
3540	17064	1102	14146	1030452
3550	17037	1102	14190	1033363
3560	17010	1102	14233	1036274
3570	16982	1102	14284	1039185
3580	16955	1102	14327	1042096
3590	16928	1102	14371	1045007
3600	16910	1102	14422	1047918
3610	16883	1102	14465	1050829
3620	16856	1102	14516	1053739
3630	16829	1102	14560	1056650
3640	16802	1102	14610	1059561
3650	16775	1102	14654	1062472
3660	16757	1102	14705	1065383
3670	16730	1102	14748	1068294
3680	16694	1102	14799	1071205
3690	16676	1102	14842	1074116
3700	16649	1102	14886	1077027
3710	16613	1102	14929	1079937
3720	16585	1102	14980	1082848
3730	16558	1102	15024	1085759
3740	16531	1102	15074	1088670
3750	16504	1102	15118	1091581
3760	16468	1102	15169	1094492
3770	16432	1102	15212	1097403
3780	16405	1102	15256	1100313

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
3790	16378	1102	15299	1103225
3800	16360	1102	15350	1106135
3810	16333	1102	15393	1109046
3820	16306	1102	15444	1111957
3830	16270	1102	15488	1114868
3840	16234	1102	15538	1117779
3850	16206	1102	15582	1120690
3860	16179	1102	15633	1123601
3870	16152	1102	15676	1126511
3880	16125	1102	15727	1129422
3890	16098	1102	15770	1132333
3900	16071	1102	15814	1135244
3910	16044	1102	15857	1138155
3920	16017	1102	15908	1141066
3930	15990	1102	15952	1143977
3940	15963	1102	16002	1146888
3950	15936	1102	16046	1149799
3960	15909	1102	16097	1152709
3970	15882	1102	16140	1155620
3980	15854	1102	16184	1158531
3990	15827	1102	16227	1161442
4000	15800	1102	16271	1164353
4010	15782	1102	16314	1167264
4020	15755	1102	16365	1170175
4030	15728	1102	16409	1173086
4040	15701	1102	16459	1175997
4050	15683	1102	16503	1178907
4060	15656	1102	16554	1181818
4070	15638	1102	16597	1184729
4080	15611	1102	16648	1187640
4090	15593	1102	16691	1190551
4100	15566	1102	16735	1193462
4110	15548	1102	16778	1196373
4120	15521	1102	16829	1199284
4130	15512	1102	16873	1202194
4140	15485	1102	16923	1205105
4150	15466	1102	16967	1208016
4160	15439	1102	17018	1210927
4170	15421	1102	17061	1213838

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
4180	15403	1102	17112	1216749
4190	15385	1102	17155	1219660
4200	15367	1102	17199	1222571
4210	15358	1102	17242	1225481
4220	15340	1102	17293	1228392
4230	15304	1102	17337	1231303
4240	15277	1102	17387	1234214
4250	15241	1102	17431	1237125
4260	15214	1102	17482	1240036
4270	15187	1102	17525	1242947
4280	15160	1102	17576	1245858
4290	15124	1102	17627	1248769
4300	15069	1102	17670	1251679
4310	15006	1102	17714	1254590
4320	14952	1102	17764	1257501
4330	14898	1102	17808	1260412
4340	14853	1102	17859	1263323
4350	14790	1102	17902	1266234
4360	14717	1102	17953	1269145
4370	14654	1102	17996	1272056
4380	14591	1102	18040	1274967
4390	14510	1102	18069	1277877
4400	14438	1102	18127	1280788
4410	14338	1102	18169	1283753
4420	14247	1102	18220	1286545
4430	14157	1102	18264	1289502
4440	14067	1102	18307	1292459
4450	13977	1102	18351	1295416
4460	13886	1102	18401	1298208
4470	13796	1102	18452	1301165
4480	13706	1102	18496	1304122
4490	13616	1102	18539	1306915
4500	13534	1102	18583	1309871
4510	13426	1102	18633	1301001
4520	13318	1102	18677	1303958
4530	13210	1102	18728	1306750
4540	13101	1102	18778	1309707
4550	12993	1102	18822	1312664
4560	12876	1102	18873	1315457

Table 2(continue).

E_b, MeV	$HL_{Qd1}, Gscm$	$HL_{Qd2}, Gscm$	$HL_{Qd3}, Gscm$	$HL_{Dip}, Gscm$
4570	12767	1102	18916	1318413
4580	12677	1102	18967	1321206
4590	12587	1102	19010	1324163
4600	12497	1102	19054	1326955
4610	12407	1102	19105	1329912
4620	12307	1102	19148	1332869
4630	12217	1102	19199	1335661
4640	12118	1102	19243	1338618
4650	12028	1102	19286	1341411
4660	11928	1102	19337	1344368
4670	11838	1102	19380	1347160
4680	11739	1102	19431	1350117
4690	11640	1102	19475	1352910
4700	11549	1102	19518	1355866
4710	11459	1102	19569	1358823
4720	11378	1102	19612	1361616
4730	11288	1102	19656	1364573
4740	11206	1102	19699	1367365
4750	11116	1102	19743	1370322
4760	11035	1102	19794	1373115
4770	10945	1102	19837	1376071
4780	10864	1102	19888	1378864
4790	10782	1102	19931	1381821
4800	10692	1102	19975	1384613
4810	10611	1102	20026	1387570
4820	10521	1102	20069	1390527
4830	10439	1102	20120	1393320
4840	10349	1102	20163	1396276
4850	10268	1102	20207	1399069
4860	10178	1102	20258	1402026
4870	10097	1102	20301	1404983
4880	10006	1102	20352	1407775
4890	9925	1102	20395	1410732
4900	9835	1102	20439	1413524
4910	9754	1102	20490	1416481
4920	9672	1102	20533	1419274
4930	9591	1102	20584	1422231
4940	9510	1102	20627	1425023
4950	9429	1102	20671	1427980

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
4960	9348	1102	20722	1430937
4970	9266	1102	20765	1433729
4980	9185	1102	20816	1436686
4990	9104	1102	20859	1439479
5000	9023	1102	20903	1442436
5010	8942	1102	20954	1445228
5020	8869	1102	20997	1448185
5030	8797	1102	21048	1451142
5040	8725	1102	21091	1453934
5050	8653	1102	21135	1456891
5060	8581	1102	21186	1459684
5070	8508	1102	21229	1462641
5080	8436	1102	21280	1465433
5090	8364	1102	21323	1468390
5100	8301	1102	21367	1471183
5110	8229	1102	21418	1474139
5120	8157	1102	21461	1477096
5130	8084	1102	21512	1479889
5140	8012	1102	21555	1482846
5150	7940	1102	21600	1485638
5160	7868	1102	21651	1488595
5170	7796	1102	21694	1491387
5180	7723	1102	21745	1494344
5190	7651	1102	21789	1497137
5200	7579	1102	21831	1500094
5210	7507	1102	21882	1503050
5220	7435	1102	21925	1505843
5230	7363	1102	21976	1508800
5240	7290	1102	22019	1511592
5250	7218	1102	22063	1514549
5260	7146	1102	22114	1517342
5270	7074	1102	22157	1520299
5280	7002	1102	22208	1523091
5290	6929	1102	22251	1526048
5300	6857	1102	22295	1529005
5310	6794	1102	22346	1531797
5320	6731	1102	22389	1534754
5330	6668	1102	22440	1537547
5340	6605	1102	22484	1540504

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
5350	6541	1102	22527	1543460
5360	6478	1102	22578	1546253
5370	6415	1102	22621	1549210
5380	6352	1102	22665	1552002
5390	6289	1102	22708	1554959
5400	6226	1102	22752	1557752
5410	6153	1102	22803	1560708
5420	6081	1102	22853	1563501
5430	6009	1102	22897	1566458
5440	5937	1102	22940	1569250
5450	5865	1102	22984	1572207
5460	5792	1102	23035	1575164
5470	5720	1102	23078	1577957
5480	5648	1102	23129	1580913
5490	5576	1102	23172	1583706
5500	5504	1102	23216	1586663
5510	5432	1102	23267	1589455
5520	5359	1102	23310	1592412
5530	5287	1102	23361	1595369
5540	5215	1102	23404	1598161
5550	5143	1102	23448	1601118
5560	5071	1102	23499	1603911
5570	4998	1102	23542	1606868
5580	4926	1102	23593	1609660
5590	4854	1102	23636	1612617
5600	4782	1102	23680	1615410
5610	4719	1102	23731	1618366
5620	4656	1102	23774	1621323
5630	4592	1102	23825	1624116
5640	4529	1102	23868	1627073
5650	4466	1102	23912	1629865
5660	4403	1102	23963	1632822
5670	4340	1102	24006	1635614
5680	4277	1102	24057	1638571
5690	4213	1102	24100	1641528
5700	4150	1102	24144	1644321
5710	4096	1102	24195	1647278
5720	4042	1102	24238	1650070
5730	3988	1102	24289	1653027

Table 2(continue).

E_b, MeV	HL_{Qd1}, G_{scm}	HL_{Qd2}, G_{scm}	HL_{Qd3}, G_{scm}	HL_{Dip}, G_{scm}
5740	3934	1102	24332	1655819
5750	3880	1102	24376	1658776
5760	3825	1102	24427	1661569
5770	3771	1102	24470	1664526
5780	3717	1102	24521	1667482
5790	3663	1102	24564	1670275
5800	3609	1102	24608	1673232
5810	3546	1102	24659	1676024
5820	3482	1102	24702	1678981
5830	3419	1102	24753	1681774
5840	3356	1102	24796	1684731
5850	3293	1102	24840	1687523
5860	3230	1102	24891	1690480
5870	3167	1102	24941	1693437
5880	3104	1102	24985	1696229
5890	3040	1102	25028	1699186
5900	2977	1102	25072	1701979
5910	2923	1102	25123	1704935
5920	2869	1102	25166	1707728
5930	2815	1102	25210	1710685
5940	2761	1102	25246	1713642
5950	2706	1102	25297	1716434
5960	2652	1102	25347	1719391
5970	2598	1102	25391	1722184
5980	2544	1102	25442	1725140
5990	2490	1102	25485	1727933
6000	2436	1102	25529	1730890

3 Appendix 1.

```

c!!!!!!!!!!!! tst.com !!!!!!!!!!!!!
#!/bin/sh
cycle=1
echo $cycle
while [ "$cycle" -le "1000" ]
do
varsetn
setqd
ray91<x.idt>x.out

```

```

setdp
echo $cycle
cycle='expr $cycle + 1'
done
exit 0

```

4 Appendix 2

```

c  !!!!!!!!!!!!!!! code varsetn.f basic !!!!!!!!!!!!!!!
    real seed
    integer*2 i1, i3, i2
    OPEN (UNIT=26,FILE='xri.out',STATUS='old')
    read(26,99), e, qd1, qd2, qd3, dip
99 format(1x,f6.1,1x,f7.4,1x,f7.4,1x,f7.4,1x,f7.4)
    close(27)
OPEN (UNIT=25,FILE='ini.out',FORM='UNFORMATTED',STATUS='UNKNOWN')
    read (25) seed
    close(25)
c Setting of magnetic field of Quads and dipole
    sqd1=qd1+0.100
    fqd1=qd1-0.100
    sqd2=qd2+0.100
    fqd2=qd2-0.100
    sqd3=qd3+0.100
    fqd3=qd3-0.100
    call rdmin(seed)
    x1=rndm(i1)
    vqd1=sqd1+x1*(fqd1-sqd1)
    x2=rndm(i2)
    vqd2=sqd2+x2*(fqd2-sqd2)
    x3=rndm(i3)
    vqd3=sqd3+x3*(fqd3-sqd3)
    vdip=dip
    OPEN (UNIT=27,FILE='xr.out',STATUS='old')
    write(27,99) e, vqd1, vqd2, vqd3, vdip
    close(27)
    call rdmout(seed)
OPEN (UNIT=25,FILE='ini.out',FORM='UNFORMATTED',STATUS='UNKNOWN')
    write(25) seed
    close(25)
end

```

5 Appendix 3

```
c !!!!!!!!!!!!! code setqd.f !!!!!!!!!!!!!!!!!!!!!!!  
    real idip  
    COMMON /KIN/ e,angle,disp  
    OPEN (UNIT=27,FILE='xr.out',STATUS='old')  
    read(27,99), e, qd1, qd2, qd3, idip  
99 format(1x,f6.1,1x,f7.4,1x,f7.4,1x,f7.4,1x,f7.4)  
    close(27)  
    OPEN (UNIT=28,FILE='x.idt',STATUS='old')  
1 format('MOLLER DETECTOR HALL A CEBAF ',f6.1,' MeV')  
    write(28,1), e  
2 format('100,500,100,2,0,0,00,')  
    write(28,2)  
3 format(f6.1,'1,1,0,0,')  
    write(28,3), e  
4 format('DRIFT')  
    write(28,4)  
5 format('67.38')  
    write(28,5)  
6 format('COLL')  
    write(28,6)  
7 format('1,0.0,0.0,4.8,4.8')  
    write(28,7)  
8 format('POLES')  
    write(28,8)  
9 format('1.693,1.693,1.693,')  
    write(28,9)  
10 format('10.16,10.16,45.24,5.08')  
    write(28,10)  
11 format(f7.4)  
    write(28,11),qd1  
12 format('5.8,-4.2,-4.2,5.8,')  
    write(28,12)  
13 format('0.1022,9.323,-1.4982,-2.1209,1.7230')  
    write(28,13)  
14 format('0.1022,9.323,-1.4982,-2.1209,1.7230')  
    write(28,14)  
15 format('1,1,1,1,')  
    write(28,15)  
16 format('COLL')  
    write(28,16)  
17 format('1,0.0,0.0,4.8,4.8')  
    write(28,17)
```

```

18 format('DRIFT')
   write(28,18)
19 format('47.34')
   write(28,19)
20 format('COLL')
   write(28,20)
21 format('1,0.0,0.0,4.8,4.8')
   write(28,21)
22 format('POLES')
   write(28,22)
23 format('1.693,1.693,1.693,')
   write(28,23)
24 format('10.16,10.16,37.02,5.08')
   write(28,24)
25 format(f6.3)
   write(28,25), qd2
26 format('5.8,-4.2,-4.2,5.8')
   write(28,26)
27 format('0.1022,8.323,-1.4982,-2.1209,1.7230')
   write(28,27)
28 format('0.1022,8.323,-1.4982,-2.1209,1.7230')
   write(28,28)
29 format('1.,1.,1.,1.,')
   write(28,29)
30 format('COLL')
   write(28,30)
31 format('1,0.0,0.0,4.8,4.8')
   write(28,31)
32 format('DRIFT')
   write(28,32)
33 format('9.05')
   write(28,33)
34 format('COLL')
   write(28,34)
35 format('1,0.0,0.0,4.8,4.8')
   write(28,35)
36 format('POLES')
   write(28,36)
37 format('1.693,1.693,1.693,')
   write(28,37)
38 format('10.16,10.16,36.34,5.08')
   write(28,38)
39 format(f6.3)
   write(28,39), qd3

```

```

40 format('5.8,-4.2,-4.2,5.8,')
   write(28,40)
41 format('0.1022,8.323,-1.4982,-2.1209,1.7230')
   write(28,41)
42 format('0.1022,8.323,-1.4982,-2.1209,1.7230')
   write(28,42)
43 format('1.,1.,1.,1.,')
   write(28,43)
44 format('COLL')
   write(28,44)
45 format('1,0.0,0.0,4.8,4.8')
   write(28,45)
46 format('DRIFT')
   write(28,46)
47 format('22.10')
   write(28,47)
48 format('COLL')
   write(28,48)
49 format('1,0.0,0.0,4.8,4.8')
   write(28,49)
50 format('COLL')
   write(28,50)
51 format('0,0.0,3.9,1.0,0.9')
   write(28,51)
52 format('DIPOLE')
   write(28,52)
53 format('1.7,1.7,1.7,1.7,0')
   write(28,53)
54 format('16.0,16.0,11.56,940.9','f6.3)
   write(28,54), idip
55 format('10.000,0.0,10.000')
   write(28,55)
56 format('0.0')
   write(28,56)
57 format('15.4,-12.6,-12.6,15.4')
   write(28,57)
58 format('0.2401,1.8369,-0.5572,0.3904')
   write(28,58)
59 format('0.2401,1.8369,-0.5572,0.3904')
   write(28,59)
60 format('0.0')
   write(28,60)
61 format('0.0')
   write(28,61)

```

```

62 format('0.0')
   write(28,62)
63 format('0.0')
   write(28,63)
64 format('COLL')
   write(28,64)
65 format('0,0.0,3.9,10.0,0.9')
   write(28,65)
66 format('DRIFT')
   write(28,66)
67 format('179.0')
   write(28,67)
68 format('SENTINEL')
   write(28,68)
   CALL ASPID
end
SUBROUTINE ASPID
real m1,m2,m3,m4
COMMON /KIN/ e,angle,disp
m1=0.511003
m2=0.511003
m3=0.511003
m4=0.511003
c Start value of T3 (degrees) in CM-system
  t3strt=90.
c Kinematic variable for moller scattering in CM-system
  w=sqrt((m1**2+m2**2+2.*m2*(e+m1)))
  e1sm=(w**2+m1**2-m2**2)/(w*2.)
  p3sm=sqrt(abs((w+m3+m4)*(w+m3-m4)*(w-m3+m4)*(w-m3-m4)))/(w*2.)
  e3sm=(w**2+m3**2-m4**2)/(w*2.)
  b3sm=p3sm/e3sm
  bsmls=sqrt(e**2-m1**2)/(e+m2)
  gamasm=(e+m2)/w
  e4sm=(w**2+m4**2-m3**2)/(2.*w)
  b4sm=p4sm/e4sm
  p1sm=gamasm*(sqrt(e**2-m1**2)-bsmls*e)
  e2sm=(w**2+m2**2-m1**2)/(2.*w)
  b2sm=p1sm/e2sm
  b1sm=p1sm/e1sm
  p2sm=p1sm
  p4sm=p3sm
  e1sk=e1sm-m1
  e2sk=e2sm-m2
  e3sk=e3sm-m3

```

```

e4sk=e4sm-m4
tet3sm=t3strt/57.29578
prom=bsmls+b3sm*cos(tet3sm)
t3lsst=atan(b3sm*sin(tet3sm)/gamasm/prom)
c Setting T3-angle in CM-system through 1 degree from 90 at CM
do 200, k=1,30
do 201, kk=1,2
ak=k-1
akk=kk
t3smgr=t3strt+ak*(-1.)**akk
tet3sm=t3smgr/57.29578
prom=bsmls+b3sm*cos(tet3sm)
tet3ls=atan(b3sm*sin(tet3sm)/gamasm/prom)
if (tet3ls.lt.0.) tet3ls=tet3ls+3.141192
t3lsgr=tet3ls*57.29578
e3ls=gamasm*(e3sm+bsmls*p3sm*cos(tet3sm))
e3ls=e3ls-m3
e4ls=e4ls-m4
c recalculation angle value in milliradian scale
angle=tet3ls*1000.
disp=((e3ls-e)*100./e)
write (28,300), angle,disp
300 format ('0.0,0.0,0.0,'1x,f7.4,',0.0,0.0,'f6.1)
201 continue
200 continue
close(28)
return
end

```

6 Appendix 4

```

x.idt-file
MOLLER DETECTOR HALL A CEBAF 1500.0 MeV
100,500,100,2,0,0,00,
1500.0,1.,1.,0.,0.,
DRIFT
67.38
COLL
1,0.0,0.0,4.8,4.8
POLES
1.693,1.693,1.693,
10.16,10.16,45.24,5.08
-0.2171

```

5.8,-4.2,-4.2,5.8,
 0.1022,9.323,-1.4982,-2.1209,1.7230
 0.1022,9.323,-1.4982,-2.1209,1.7230
 1.,1.,1.,1.,
 COLL
 1,0.0,0.0,4.8,4.8
 DRIFT
 47.34
 COLL
 1,0.0,0.0,4.8,4.8
 POLES
 1.693,1.693,1.693,
 10.16,10.16,37.02,5.08
 -0.015
 5.8,-4.2,-4.2,5.8
 0.1022,8.323,-1.4982,-2.1209,1.7230
 0.1022,8.323,-1.4982,-2.1209,1.7230
 1.,1.,1.,1.,
 COLL
 1,0.0,0.0,4.8,4.8
 DRIFT
 9.05
 COLL
 1,0.0,0.0,4.8,4.8
 POLES
 1.693,1.693,1.693,
 10.16,10.16,36.34,5.08
 -0.065
 5.8,-4.2,-4.2,5.8,
 0.1022,8.323,-1.4982,-2.1209,1.7230
 0.1022,8.323,-1.4982,-2.1209,1.7230
 1.,1.,1.,1.,
 COLL
 1,0.0,0.0,4.8,4.8
 DRIFT
 22.10
 COLL
 1,0.0,0.0,4.8,4.8
 COLL
 0,0.0,3.9,1.0,0.9
 DIPOLE
 1.7,1.7,1.7,1.7,0
 16.0,16.0,11.56,940.9, 0.246
 10.000,0.0,10.000

0.0
 15.4,-12.6,-12.6,15.4
 0.2401,1.8369,-0.5572,0.3904
 0.2401,1.8369,-0.5572,0.3904
 0.0
 0.0
 0.0
 0.0
 COLL
 0,0.0,3.9,10.0,0.9
 DRIFT
 179.0
 SENTINEL
 0.0,0.0,0.0, 26.0965,0.0,0.0, -50.0
 0.0,0.0,0.0, 26.0965,0.0,0.0, -50.0
 0.0,0.0,0.0, 25.6452,0.0,0.0, -49.1
 0.0,0.0,0.0, 26.5558,0.0,0.0, -50.9
 0.0,0.0,0.0, 25.2015,0.0,0.0, -48.3
 0.0,0.0,0.0, 27.0233,0.0,0.0, -51.8
 0.0,0.0,0.0, 24.7652,0.0,0.0, -47.4
 0.0,0.0,0.0, 27.4993,0.0,0.0, -52.6
 0.0,0.0,0.0, 24.3361,0.0,0.0, -46.5
 0.0,0.0,0.0, 27.9841,0.0,0.0, -53.5
 0.0,0.0,0.0, 23.9139,0.0,0.0, -45.7
 0.0,0.0,0.0, 28.4781,0.0,0.0, -54.4
 0.0,0.0,0.0, 23.4984,0.0,0.0, -44.8
 0.0,0.0,0.0, 28.9816,0.0,0.0, -55.2
 0.0,0.0,0.0, 23.0894,0.0,0.0, -43.9
 0.0,0.0,0.0, 29.4949,0.0,0.0, -56.1
 0.0,0.0,0.0, 22.6866,0.0,0.0, -43.1
 0.0,0.0,0.0, 30.0184,0.0,0.0, -57.0
 0.0,0.0,0.0, 22.2899,0.0,0.0, -42.2
 0.0,0.0,0.0, 30.5525,0.0,0.0, -57.8
 0.0,0.0,0.0, 21.8990,0.0,0.0, -41.3
 0.0,0.0,0.0, 31.0977,0.0,0.0, -58.7
 0.0,0.0,0.0, 21.5139,0.0,0.0, -40.5
 0.0,0.0,0.0, 31.6542,0.0,0.0, -59.6
 0.0,0.0,0.0, 21.1342,0.0,0.0, -39.6
 0.0,0.0,0.0, 32.2227,0.0,0.0, -60.4
 0.0,0.0,0.0, 20.7598,0.0,0.0, -38.8
 0.0,0.0,0.0, 32.8035,0.0,0.0, -61.3
 0.0,0.0,0.0, 20.3906,0.0,0.0, -37.9
 0.0,0.0,0.0, 33.3972,0.0,0.0, -62.1
 0.0,0.0,0.0, 20.0264,0.0,0.0, -37.1

0.0,0.0,0.0, 34.0043,0.0,0.0, -63.0
 0.0,0.0,0.0, 19.6671,0.0,0.0, -36.2
 0.0,0.0,0.0, 34.6253,0.0,0.0, -63.8
 0.0,0.0,0.0, 19.3124,0.0,0.0, -35.4
 0.0,0.0,0.0, 35.2608,0.0,0.0, -64.6
 0.0,0.0,0.0, 18.9623,0.0,0.0, -34.6
 0.0,0.0,0.0, 35.9115,0.0,0.0, -65.5
 0.0,0.0,0.0, 18.6165,0.0,0.0, -33.7
 0.0,0.0,0.0, 36.5780,0.0,0.0, -66.3
 0.0,0.0,0.0, 18.2751,0.0,0.0, -32.9
 0.0,0.0,0.0, 37.2609,0.0,0.0, -67.1
 0.0,0.0,0.0, 17.9378,0.0,0.0, -32.1
 0.0,0.0,0.0, 37.9610,0.0,0.0, -67.9
 0.0,0.0,0.0, 17.6045,0.0,0.0, -31.3
 0.0,0.0,0.0, 38.6792,0.0,0.0, -68.7
 0.0,0.0,0.0, 17.2751,0.0,0.0, -30.5
 0.0,0.0,0.0, 39.4161,0.0,0.0, -69.5
 0.0,0.0,0.0, 16.9495,0.0,0.0, -29.7
 0.0,0.0,0.0, 40.1726,0.0,0.0, -70.3
 0.0,0.0,0.0, 16.6276,0.0,0.0, -28.9
 0.0,0.0,0.0, 40.9497,0.0,0.0, -71.1
 0.0,0.0,0.0, 16.3092,0.0,0.0, -28.1
 0.0,0.0,0.0, 41.7484,0.0,0.0, -71.9
 0.0,0.0,0.0, 15.9942,0.0,0.0, -27.3
 0.0,0.0,0.0, 42.5696,0.0,0.0, -72.7
 0.0,0.0,0.0, 15.6826,0.0,0.0, -26.6
 0.0,0.0,0.0, 43.4145,0.0,0.0, -73.5
 0.0,0.0,0.0, 15.3743,0.0,0.0, -25.8
 0.0,0.0,0.0, 44.2842,0.0,0.0, -74.2

7 Appendix 5

```

c !!!!!!!!!!!!!!! code setdp.f !!!!!!!!!!!!!!!
  PROGRAM setdp
  character*1 aaa(130)
  c determination of number of acceptable trajectories from Raytrace out-file
  c rewriting out-files of RAYTRACE
  c the all rewritings are started from determination of first line number
  c the begining of rewriting upto "1" at 5-th column of Raytrace out-file
  c reading of Raytrace x.out-file from 1-th ray upto 100-th ray
    OPEN (UNIT=18,FILE='x.out',STATUS='old')
    do 2 i=1,1000
      read(18,1), ii

```

```

    ni=i-1
1  format (2x,I3)
    if (ii.eq.1) go to 3
2  continue
3  continue
    close(18)
c exit with determined number (ni+1) the first trajectory line
c reading upto the first trajectory line without rewriting
    nif=ni+119
    OPEN (UNIT=29,FILE='xtoy.out',STATUS='old')
    OPEN (UNIT=18,FILE='x.out',STATUS='old')
    do 4 i=1,nif
        read(18,111) aaa
111 format(130a1)
        if (i.gt.ni) go to 5
        go to 4
c after this - rewriting to new file
    5 continue
        write(29,111) aaa
    4 continue
        close(18)
        close(29)
c looking for the first stopping trajectory
c the looking for ***-lines of out-file
    OPEN (UNIT=29,FILE='xtoy.out',STATUS='old')
    do 7 j=1,119
        read (29,111) aaa
        jf=(j-1)/2
        do 8 i=1,87
            if (aaa(i).eq.'*') go to 9
8  continue
7  continue
9  continue
    print *,jf='jf'
    close(29)
    OPEN (UNIT=27,FILE='xr.out',STATUS='old')
    OPEN (UNIT=30,FILE='xf.out',STATUS='old')
    read(27,110), e, qd1, qd2, qd3, dip
    write(30,112), e, qd1, qd2, qd3, dip, jf
    close(27)
    close(30)
110 format(1x,f6.1,1x,f7.4,1x,f7.4,1x,f7.4,1x,f7.4)
112 format(1x,f6.1,1x,f7.4,1x,f7.4,1x,f7.4,1x,f7.4,1x,i3)
    OPEN (UNIT=30,FILE='xf.out',STATUS='old')

```

```

      OPEN (UNIT=31,FILE='xfa.out',STATUS='unknown')
200 continue
      read(31,112,err=201), e, qd1, qd2, qd3, dip, jf
      go to 200
201 continue
202 continue
      read(30,112,err=203), e, qd1, qd2, qd3, dip, jf
      write(31,112,err=204), e, qd1, qd2, qd3, dip, jf
      go to 202
203 continue
204 continue
      close(30)
      close(31)
      end

```

8 Appendix 6

```

c !!!!!!!!!!!!!!!! code varsetns.f basic !!!!!!!!!!!!!!!!
      real seed
      integer*2 i1
      OPEN (UNIT=26,FILE='xri.out',STATUS='old')
      read(26,99), e, qd1, qd2, qd3, dip
99 format(1x,f6.1,1x,f7.4,1x,f7.4,1x,f7.4,1x,f7.4)
      close(27)
OPEN (UNIT=25,FILE='ini.out',FORM='UNFORMATTED',STATUS='UNKNOWN')
      read (25) seed
      close(25)
c Setting of magnetic field of Quads and dipole
      sqd1=qd1+0.100
      fqd1=qd1-0.100
      sqd2=qd2
      sqd3=qd3
      call rdmin(seed)
      x1=rndm(i1)
      vqd1=sqd1+x1*(fqd1-sqd1)
      vqd2=sqd2
      vqd3=sqd3
      vdip=dip
      OPEN (UNIT=27,FILE='xr.out',STATUS='old')
      write(27,99) e, vqd1, vqd2, vqd3, vdip
      close(27)
      call rdmout(seed)
OPEN (UNIT=25,FILE='ini.out',FORM='UNFORMATTED',STATUS='UNKNOWN')

```

```
write(25) seed  
close(25)  
end
```